

Python Environment Comparison: uv vs venv vs Conda

Overview

Python offers several tools for managing environments. This cheat sheet compares uv, venv, and Conda to help choose the right tool for your project.

venv

- Built into Python
- Lightweight and simple
- Uses pip for dependency management
- Ideal for most standard Python projects

uv

- Third-party tool focused on speed
- Drop-in replacement for pip + venv
- Extremely fast dependency resolution
- Python-only environments

Conda

- Full environment and package manager
- Manages Python and non-Python dependencies
- Heavier but powerful
- Common in data science and ML workflows

Key Differences

- venv is built-in; uv and Conda must be installed
- uv focuses on speed; Conda focuses on completeness
- Conda can manage system-level libraries
- uv environments are smaller and faster than Conda

When to Use Each

Use venv when:

- You want simplicity and minimal tooling

Use uv when:

- You want fast installs and modern workflows

Use Conda when:

- You need non-Python dependencies or scientific packages

Team & Project Recommendations

- Do not mix tools within a single project
- Document environment setup clearly
- Choose uv or Conda for reproducibility in teams